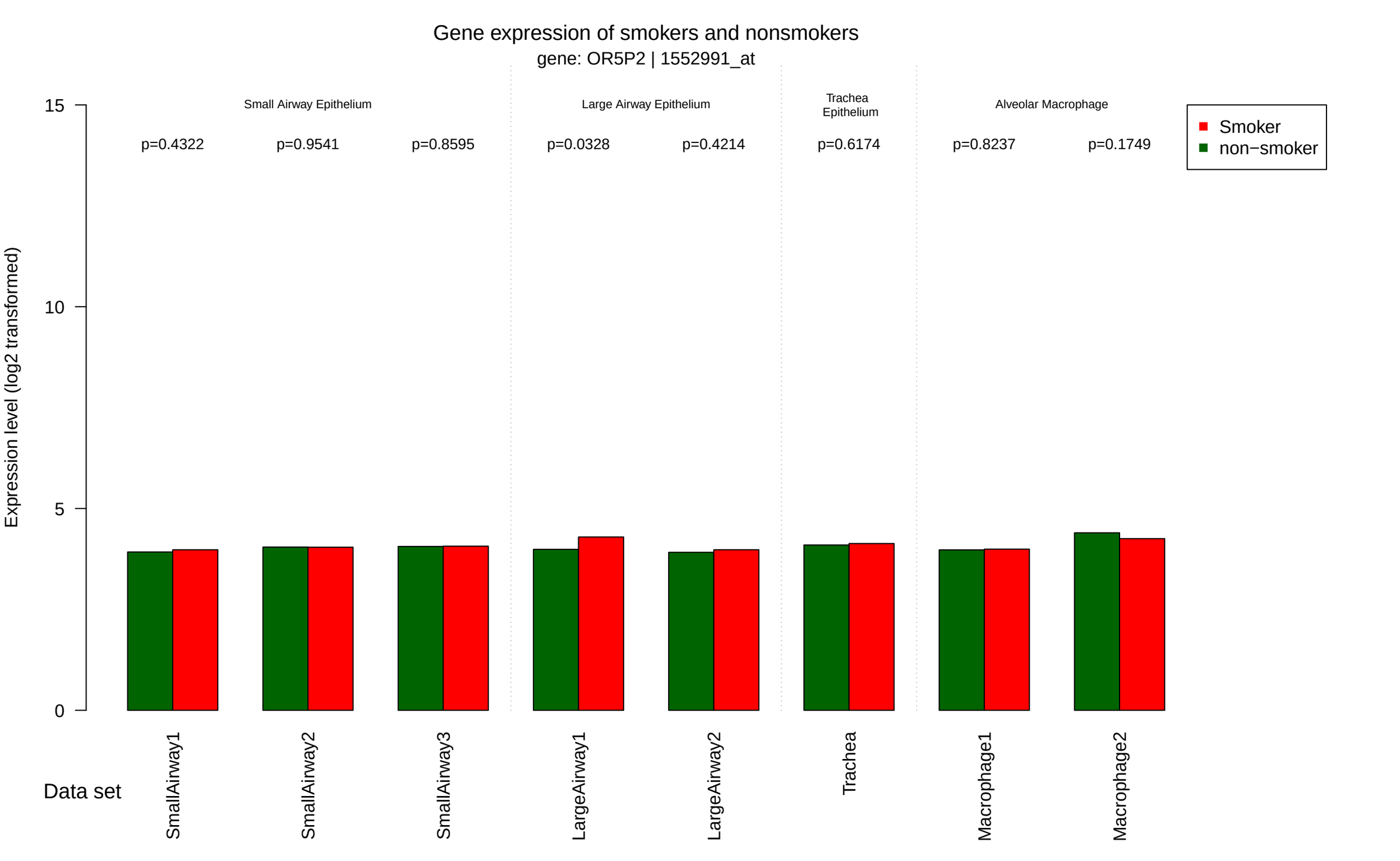
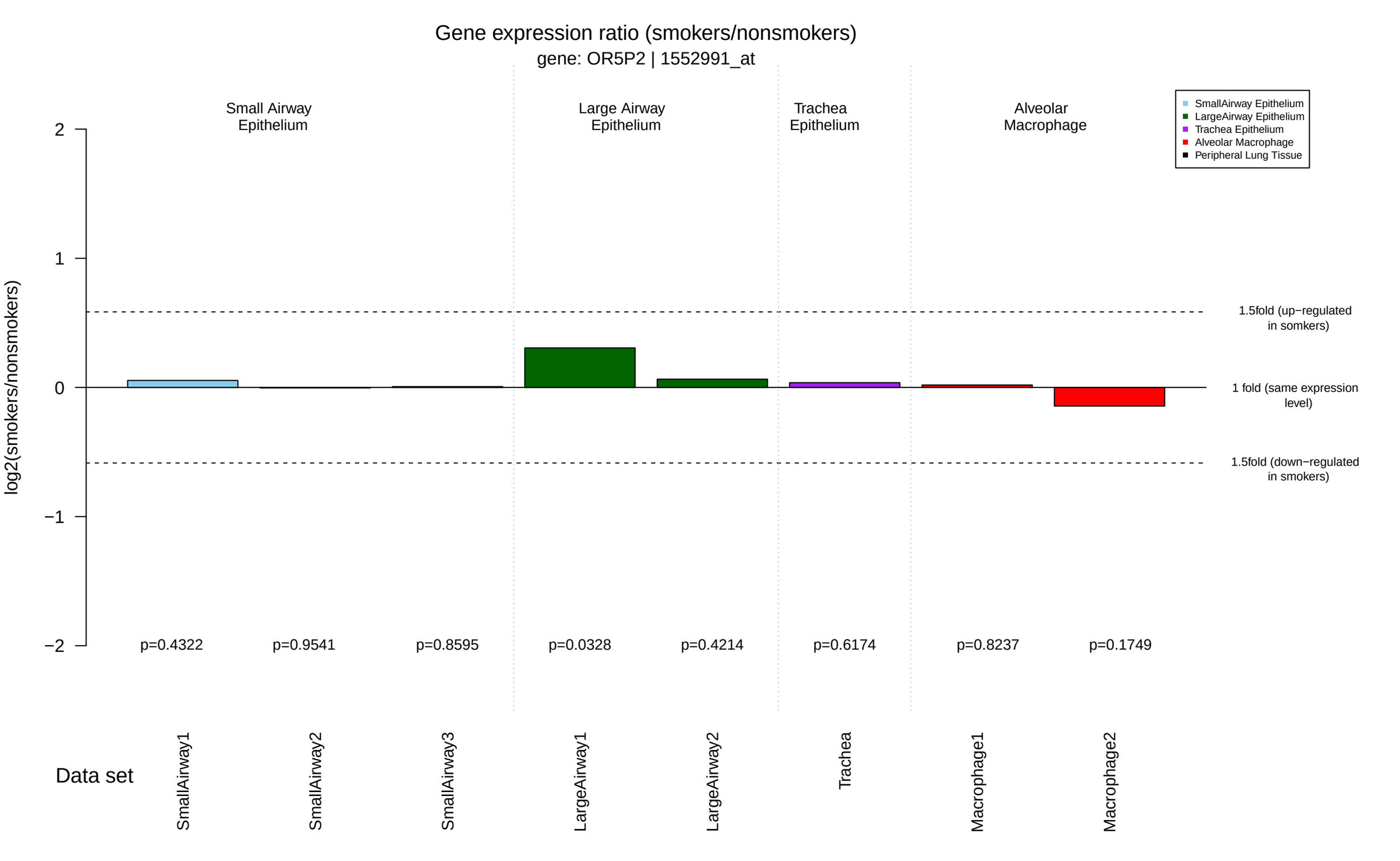
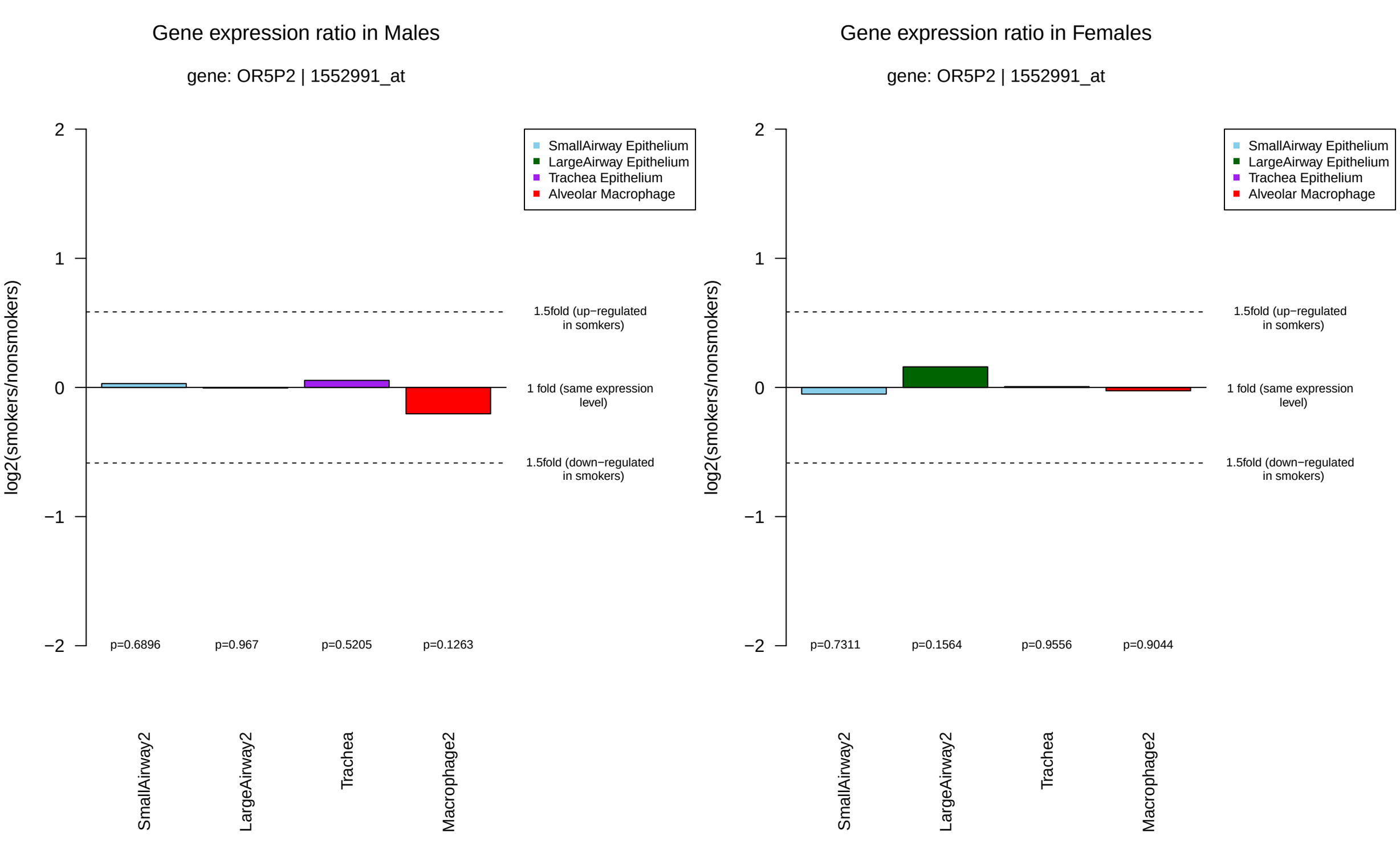
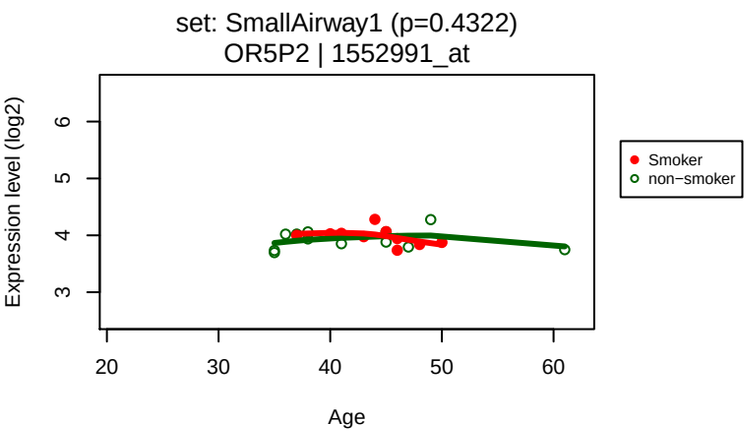
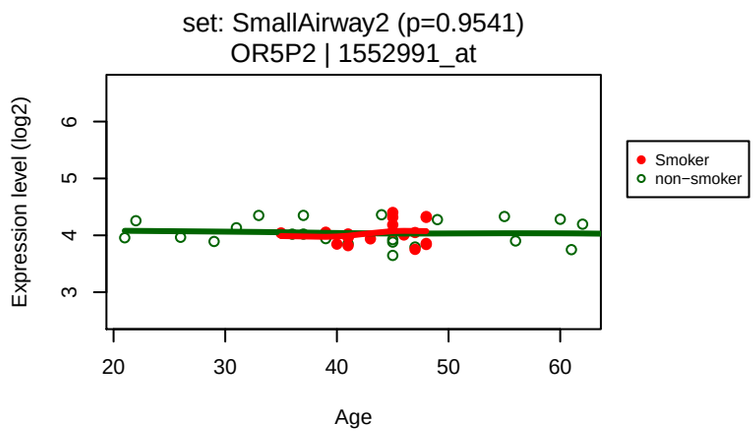
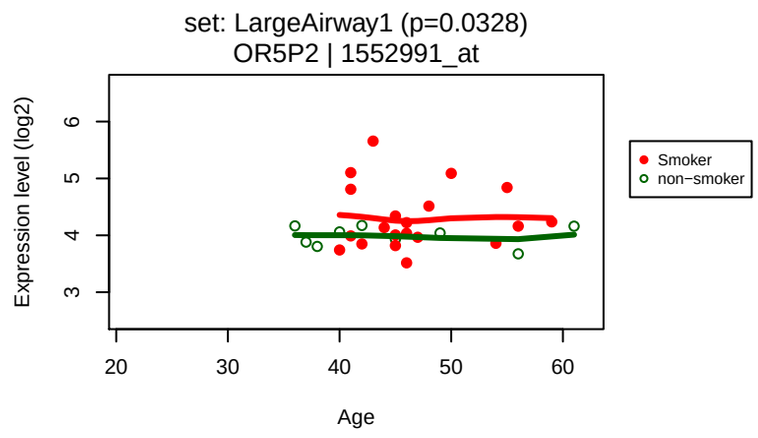

Gene expression of smokers and nonsmokers
gene: OR5P2 | 1552991_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4322
p=0.9541
p=0.8595
p=0.0328
p=0.4214
p=0.6174
p=0.8237
p=0.1749
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

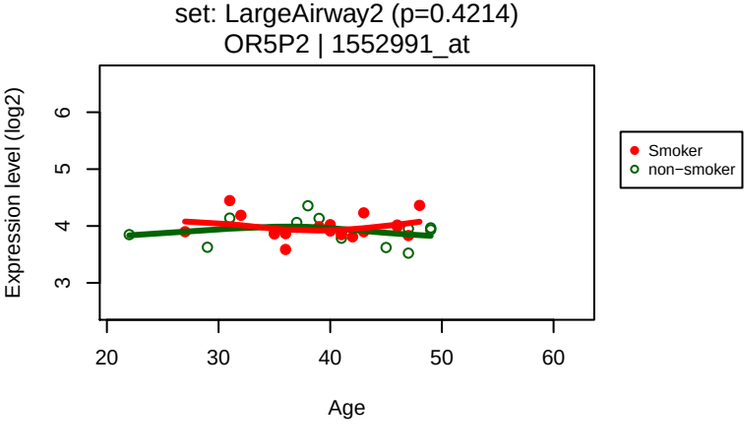

Gene expression ratio (smokers/nonsmokers)
gene: OR5P2 | 1552991_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4322
p=0.9541
p=0.8595
p=0.0328
p=0.4214
p=0.6174
p=0.8237
p=0.1749
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

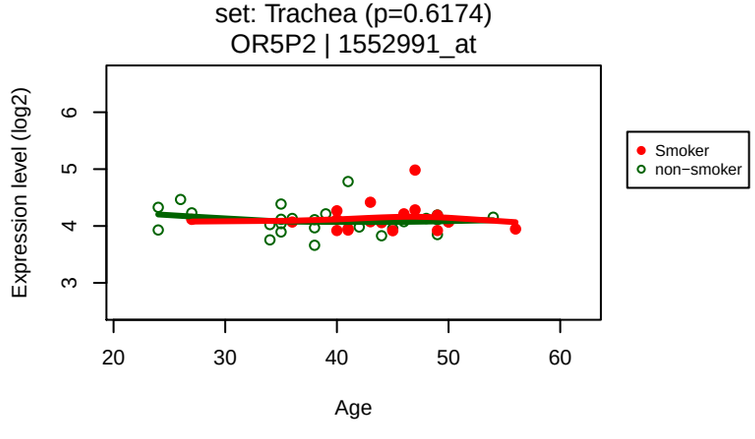

Gene expression ratio in Males
gene: OR5P2 | 1552991_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6896
p=0.967
p=0.5205
p=0.1263
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OR5P2 | 1552991_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7311
p=0.1564
p=0.9556
p=0.9044
SmallAirway2
LargeAirway2
Trachea
Macrophage2

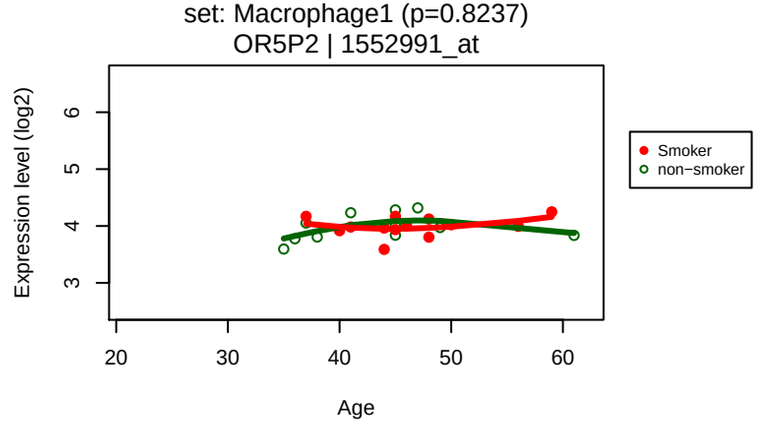

set: SmallAirway1 (p=0.4322)
OR5P2 | 1552991_at
Expression level (log2)
Age
Smoker
non−smoker

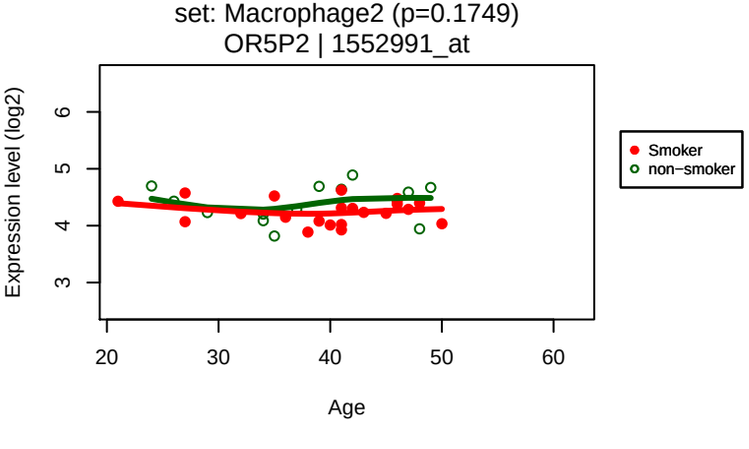

set: SmallAirway2 (p=0.9541)
OR5P2 | 1552991_at
Expression level (log2)
Age
Smoker
non−smoker

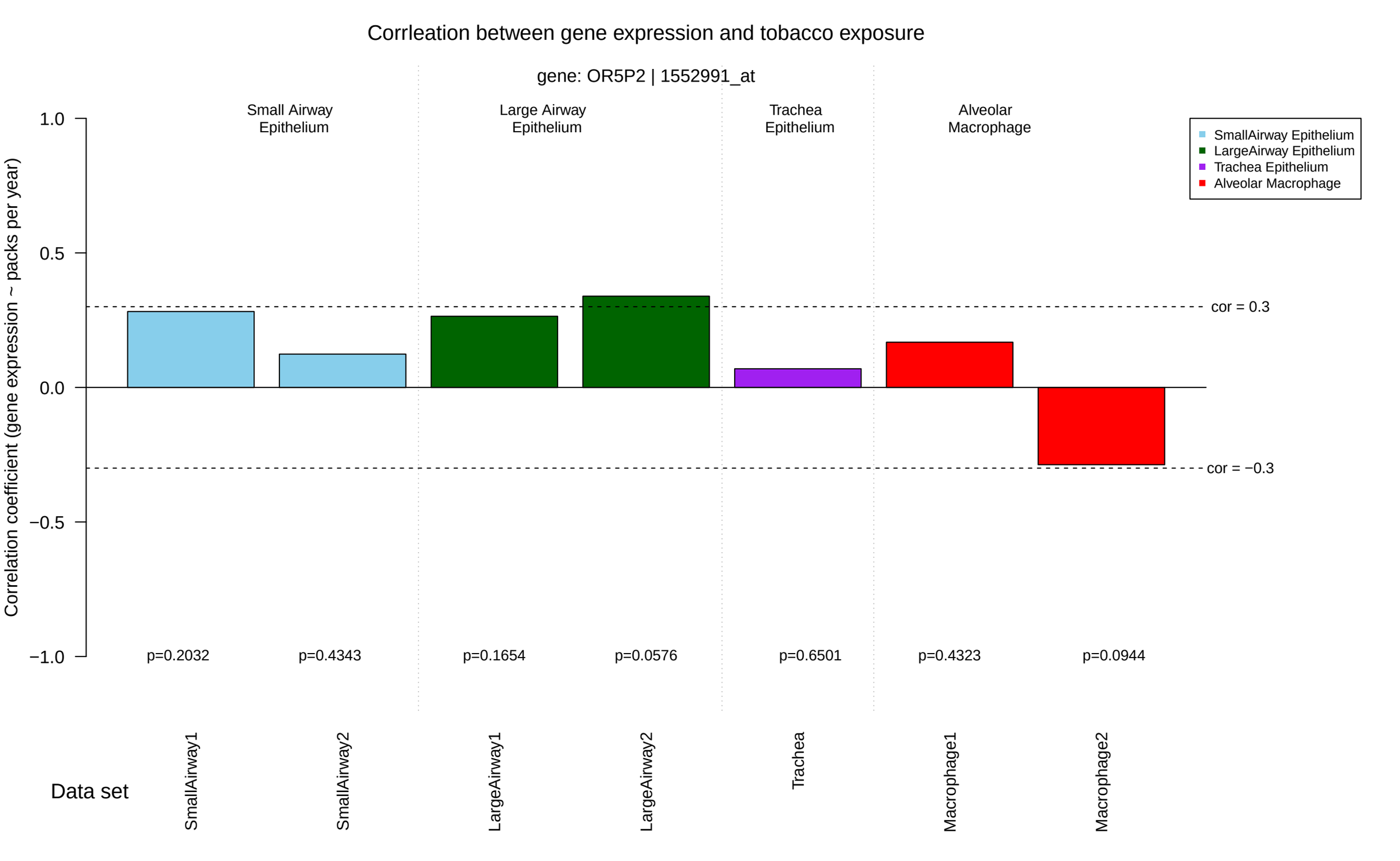

set: LargeAirway1 (p=0.0328)
OR5P2 | 1552991_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4214)
OR5P2 | 1552991_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6174)
OR5P2 | 1552991_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8237)
OR5P2 | 1552991_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1749)
OR5P2 | 1552991_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OR5P2 | 1552991_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2032
p=0.4343
p=0.1654
p=0.0576
p=0.6501
p=0.4323
p=0.0944
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2